

## DOLE-JURA AP, France

WMO: 073860

Lat: 47.0439N

Lon: 5.4272E

Elev: 640

StdP: 14.36

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 19.8       | 23.3 | 12.7                          | 10.7 | 23.1 | 18.1 | 14.0 | 25.8 | 31.4                  | 50.0 | 27.2 | 46.0 | 7.0                   | 20   | 0.518 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 21.7                   | 90.4            | 69.5 | 86.9 | 68.6 | 83.6 | 67.6 | 71.5                | 85.2 | 70.0 | 83.4 | 68.7 | 80.7 | 7.9                  | 200  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 67.3                            | 102.8 | 76.4 | 65.8 | 97.5 | 74.1 | 64.6 | 93.2 | 72.9 | 35.6          | 85.0 | 34.3 | 83.6 | 33.2 | 80.7 | 77.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |      |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |      |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max  |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p)  |       |
| 25.9              | 21.5 | 18.4 | DB                         | 14.1 | 95.7               | 5.8 | 3.6                                                | 9.9 | 98.3       | 6.5 | 100.4      | 3.2 | 102.4      | -1.0 | 105.0 |
|                   |      |      | WB                         | 13.4 | 74.4               | 5.9 | 1.3                                                | 9.2 | 75.3       | 5.8 | 76.1       | 2.5 | 76.8       | -1.8 | 77.8  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 52.9        | 37.3  | 38.7  | 45.6  | 52.1  | 57.9  | 66.3  | 69.3  | 67.9  | 61.8  | 54.4  | 44.4  | 38.5  |
|                                                                        | DBStd       | 13.15       | 7.62  | 7.04  | 6.91  | 6.06  | 5.97  | 6.12  | 5.82  | 5.75  | 5.54  | 6.88  | 6.87  | 7.78  |
|                                                                        | HDD50       | 1527        | 396   | 317   | 169   | 47    | 6     | 0     | 0     | 0     | 0     | 38    | 193   | 361   |
|                                                                        | HDD65       | 4825        | 858   | 735   | 602   | 386   | 231   | 58    | 24    | 33    | 125   | 331   | 619   | 823   |
|                                                                        | CDD50       | 2601        | 4     | 2     | 32    | 111   | 252   | 489   | 598   | 556   | 355   | 175   | 24    | 3     |
|                                                                        | CDD65       | 420         | 0     | 0     | 0     | 0     | 12    | 96    | 156   | 123   | 30    | 3     | 0     | 0     |
|                                                                        | CDH74       | 4531        | 0     | 0     | 0     | 40    | 183   | 1010  | 1719  | 1264  | 297   | 18    | 0     | 0     |
|                                                                        | CDH80       | 1576        | 0     | 0     | 0     | 1     | 25    | 318   | 691   | 484   | 57    | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 7.6         | 8.6   | 9.1   | 9.1   | 7.9   | 7.4   | 6.8   | 6.3   | 6.0   | 6.5   | 6.9   | 7.9   | 8.5   |
| (15) Precipitation                                                     | PrecAvg     | 34.4        | 2.4   | 2.2   | 2.2   | 2.7   | 3.3   | 2.9   | 3.1   | 2.9   | 2.8   | 3.5   | 3.7   | 2.7   |
|                                                                        | PrecMax     | 43.6        | 5.7   | 4.3   | 8.5   | 6.3   | 6.6   | 5.7   | 6.1   | 5.2   | 5.8   | 7.8   | 10.1  | 5.0   |
|                                                                        | PrecMin     | 26.2        | 0.6   | 0.1   | 0.2   | 0.0   | 0.6   | 0.5   | 0.9   | 0.6   | 0.1   | 0.7   | 1.2   | 0.2   |
|                                                                        | PrecStd     | 4.6         | 1.3   | 1.2   | 1.7   | 1.6   | 1.5   | 1.3   | 1.4   | 1.3   | 1.7   | 1.7   | 2.1   | 1.3   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 56.6  | 59.8  | 69.4  | 78.9  | 83.6  | 91.6  | 95.9  | 97.2  | 85.3  | 76.6  | 64.8  | 58.0  |
|                                                                        |             | MCWB        | 51.4  | 49.8  | 53.9  | 61.3  | 66.0  | 69.7  | 69.3  | 69.9  | 66.8  | 62.6  | 57.5  | 52.2  |
|                                                                        | 2%          | DB          | 53.7  | 56.0  | 65.8  | 73.9  | 78.8  | 87.4  | 90.9  | 89.9  | 81.4  | 72.4  | 60.9  | 54.7  |
|                                                                        |             | MCWB        | 50.0  | 48.3  | 52.9  | 58.7  | 63.7  | 70.0  | 69.4  | 69.6  | 66.2  | 61.9  | 55.3  | 50.6  |
|                                                                        | 5%          | DB          | 51.4  | 52.6  | 61.9  | 69.7  | 75.3  | 83.7  | 87.4  | 85.6  | 77.4  | 68.8  | 57.8  | 52.2  |
|                                                                        |             | MCWB        | 48.2  | 46.9  | 51.1  | 56.6  | 62.3  | 68.5  | 68.4  | 68.5  | 64.8  | 60.6  | 53.7  | 49.3  |
|                                                                        | 10%         | DB          | 48.4  | 49.8  | 58.0  | 65.7  | 71.2  | 80.2  | 83.8  | 81.3  | 73.7  | 65.4  | 54.9  | 49.7  |
|                                                                        |             | MCWB        | 45.8  | 45.7  | 49.8  | 54.4  | 60.1  | 66.4  | 67.1  | 66.9  | 62.9  | 59.1  | 51.9  | 47.5  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 52.8  | 51.5  | 55.8  | 62.7  | 68.6  | 73.6  | 72.7  | 73.5  | 69.6  | 65.3  | 58.1  | 53.3  |
|                                                                        |             | MCDB        | 55.3  | 56.9  | 65.4  | 76.3  | 79.5  | 85.8  | 87.6  | 86.9  | 80.4  | 72.7  | 62.7  | 56.3  |
|                                                                        | 2%          | WB          | 50.3  | 49.4  | 53.8  | 59.6  | 65.7  | 71.5  | 71.0  | 71.0  | 67.6  | 63.2  | 56.0  | 51.1  |
|                                                                        |             | MCDB        | 53.0  | 54.4  | 62.6  | 71.2  | 75.8  | 83.5  | 85.7  | 85.0  | 78.2  | 70.0  | 59.6  | 53.8  |
|                                                                        | 5%          | WB          | 48.4  | 47.6  | 52.2  | 57.4  | 63.4  | 69.6  | 69.6  | 69.4  | 65.7  | 61.3  | 54.1  | 49.5  |
|                                                                        |             | MCDB        | 51.2  | 51.8  | 59.9  | 68.3  | 72.4  | 81.3  | 84.1  | 82.5  | 75.1  | 67.0  | 57.3  | 52.0  |
|                                                                        | 10%         | WB          | 46.1  | 45.8  | 50.5  | 55.2  | 61.2  | 67.7  | 68.2  | 68.0  | 63.9  | 59.6  | 52.0  | 47.5  |
|                                                                        |             | MCDB        | 48.2  | 49.6  | 57.4  | 64.1  | 69.0  | 78.1  | 81.0  | 78.4  | 71.9  | 64.6  | 54.8  | 49.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 9.8   | 13.8  | 18.2  | 20.2  | 19.5  | 20.8  | 21.7  | 20.7  | 19.5  | 16.5  | 11.6  | 9.9   |
|                                                                        |             | MCDBR       | 11.7  | 20.1  | 26.2  | 28.7  | 26.9  | 26.9  | 28.7  | 28.6  | 25.6  | 21.6  | 15.9  | 13.2  |
|                                                                        | 5% WB       | MCWBR       | 8.7   | 13.9  | 15.1  | 15.3  | 14.1  | 11.8  | 10.1  | 11.1  | 12.3  | 12.4  | 11.2  | 10.1  |
|                                                                        |             | MCWBR       | 10.9  | 17.9  | 22.3  | 26.8  | 23.8  | 25.1  | 25.7  | 24.6  | 22.1  | 18.4  | 14.3  | 12.7  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.312 | 0.333 | 0.361 | 0.396 | 0.402 | 0.409 | 0.405 | 0.399 | 0.377 | 0.363 | 0.330 | 0.309 |
|                                                                        |             | taud        | 2.455 | 2.399 | 2.338 | 2.251 | 2.280 | 2.297 | 2.304 | 2.335 | 2.376 | 2.410 | 2.452 | 2.455 |
|                                                                        | Ebn at Noon |             | 247   | 264   | 272   | 272   | 274   | 272   | 271   | 268   | 264   | 249   | 236   | 233   |
|                                                                        |             | Edh at Noon | 21    | 27    | 34    | 40    | 41    | 40    | 40    | 37    | 32    | 27    | 21    | 19    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 340   | 602   | 1007  | 1406  | 1644  | 1921  | 1836  | 1590  | 1228  | 754   | 399   | 278   |
|                                                                        | RadStd      |             | 44    | 97    | 135   | 195   | 175   | 177   | 158   | 135   | 99    | 61    | 38    | 46    |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (63 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page